

MUHAMMAD USMAN BARI

+92 325 1053811 | usmanbari2005@gmail.com | Islamabad, Pakistan

PROFESSIONAL SUMMARY

Computer Science undergraduate at FAST-NUCES with a strong foundation in programming, software engineering, and systems thinking, complemented by hands-on experience in automation, scripting, and full-stack development. Comfortable working with Python, C++, and modern development tools, with coursework spanning data structures, operating systems, database systems, and intelligent systems. Eager to apply strong analytical and problem-solving skills to operations automation, monitoring, and process optimization in a structured, enterprise engineering environment.

EDUCATION

FAST-NUCES, Islamabad

August 2023 - Present

BS Computer Science

- Relevant Coursework: Operating Systems, Database Systems, Data Structures & Algorithms, Object-Oriented Programming, Introduction to Artificial Intelligence
- Coursework covered automation-relevant concepts including process scheduling, system resource management, algorithmic problem-solving, and structured data handling

EXPERIENCE

Frontend Web Development Intern - CodeAlpha

July 2025

- Collaborated within a development team using version control (Git) and structured workflows, contributing to code reviews and iterative feature delivery in a collaborative, process-driven environment
- Built and optimized component-based interfaces with a focus on reliability, maintainability, and cross-device performance, following clean coding and documentation best practices
- Worked with React, TypeScript, HTML, and CSS to deliver responsive, accessible applications under team coding standards

PROJECTS

XONIX Arcade Game (C++)

May 2025

- Engineered real-time collision detection and dynamic region-filling logic, applying algorithmic problem-solving and efficient data structure management to handle live system state
- Built and debugged a real-time system using SFML for rendering, demonstrating strong systems programming fundamentals

AI-Powered Conversational Assistant

March 2025

- Developed a full-stack application integrating an external API to process requests and return real-time, context-aware responses
- Designed a modular, reusable architecture for maintainability and ease of monitoring/extension - directly relevant to building automation tools and internal dashboards

Notes App

September 2025

- Built a full CRUD application with persistent state management, focusing on clean, scalable, and maintainable code structure

Retro Gameboy Application (C++/SFML)

December 2024

- Applied Object-Oriented Programming principles to design a real-time interactive system handling continuous input and state updates

CERTIFICATIONS

- Google Data Analytics Professional Certificate - Coursera (May 2025)
- Meta Front-End Developer Professional Certificate - Coursera (April 2025)

SKILLS

- Programming: Python, C++, JavaScript, TypeScript
- Automation & Data: Python scripting, NumPy, Pandas, Scikit-learn, data analysis fundamentals
- Tools & Platforms: Git, Linux, VS Code, Windows
- Web Technologies: ReactJS, Next.js, HTML5, CSS3, Tailwind CSS
- Core CS: Data Structures & Algorithms, OOP, Operating Systems, Database Systems